

A PROJECT TITLE
ON
BANKRUPTCY PREDICTION USING DECISION TREES AND RANDOM FOREST
WITH PRECISION-RECALL ANALYSIS

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING
BACHELOR OF TECHNOLOGY

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by

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DEPARTMENT OF CSIT

Name of Title: Bankruptcy Prediction Using Decision Trees and Random Forest with Precision-Recall Analysis

1. Abstract:

Bankruptcy prediction is a critical task in financial risk management, as the timely identification of financially distressed companies allows banks, investors, lenders, companies, and regulatory authorities to take corrective action before irreversible losses occur. This project proposes a machine-learning-based system to predict whether a company is likely to become bankrupt by analysing a range of company financial indicators using supervised classification algorithms. Two algorithms are proposed for this task: the Decision Tree Classifier, which produces an interpretable model by partitioning the financial feature space using threshold-based rules, and the Random Forest Classifier, which combines the predictions of multiple decision trees to improve generalisation and reduce the risk of overfitting.

Before model training, the dataset will undergo systematic data preprocessing, including handling of missing values, removal of unnecessary or redundant columns, and appropriate preparation of numerical features. The processed dataset will then be split into training and testing sets to evaluate model performance on unseen data. A significant challenge associated with bankruptcy datasets is class imbalance, since the number of non-bankrupt companies is typically far greater than the number of bankrupt companies. Under such conditions, accuracy alone is not a reliable evaluation metric, as a model may achieve a high accuracy score while still failing to correctly identify bankrupt companies. For this reason, the project places emphasis on precision, recall, and F1-score, together with the confusion matrix, as evaluation measures. The Precision-Recall curve and the Average Precision score will also be used, as they are particularly suitable for evaluating classifiers on imbalanced datasets.

In addition, feature-importance analysis will be carried out to identify which financial indicators contribute most significantly to the prediction of bankruptcy. The performance of the Decision Tree and Random Forest models will be compared to determine which model provides a better balance between correctly detecting bankrupt companies and minimising incorrect bankruptcy predictions. The resulting system is intended to function as a decision-support tool for financial-risk assessment and the early detection of financial distress, thereby assisting stakeholders in making informed and timely financial decisions.

2. INTRODUCTION:

Bankruptcy refers to a legal condition in which a company is unable to meet its financial obligations and repay its outstanding debts to creditors. It generally arises as the final stage of a longer process known as financial distress, during which a company experiences declining profitability, reduced liquidity, and increasing difficulty in meeting its short-term and long-term financial commitments. Identifying signs of financial distress at an early stage is of considerable importance, since it allows companies, banks, investors, and regulatory authorities to take timely corrective measures, such as restructuring debt, adjusting credit exposure, or revising investment decisions, before the situation deteriorates into full-scale bankruptcy.

Traditionally, the financial health of a company has been assessed through manual analysis of financial statements and ratios by financial analysts. However, manual analysis has notable limitations: it is time-consuming, dependent on the expertise of the analyst, and often unable to process the large volume of financial data generated by modern companies. Moreover, manual methods may fail to capture complex, non-linear relationships between financial indicators that influence the likelihood of bankruptcy. Machine

learning offers an alternative approach, as it can automatically learn patterns from historical financial data and use these patterns to classify companies as bankrupt or non-bankrupt.

Among the various machine-learning techniques available for classification tasks, the Decision Tree Classifier is one of the most intuitive. It works by repeatedly splitting the dataset based on the values of financial indicators, forming a tree-like structure in which each internal node represents a decision rule and each leaf node represents a predicted class. While Decision Trees are easy to interpret, they can be prone to overfitting, particularly when the tree is allowed to grow without restriction. The Random Forest Classifier addresses this limitation by constructing a large number of decision trees, each trained on a different random subset of the data and features, and combining their individual predictions through a voting mechanism. This ensemble approach generally improves predictive performance and reduces overfitting compared to a single decision tree.

A key characteristic of bankruptcy datasets is class imbalance, which occurs because the number of companies that actually go bankrupt is typically much smaller than the number of companies that remain financially stable. When a dataset is imbalanced, a classifier can achieve a misleadingly high accuracy simply by predicting the majority class for most instances, while performing poorly on the minority class, which in this case represents bankrupt companies. This makes accuracy an unreliable metric for evaluating model performance. Precision, which measures the proportion of predicted bankrupt companies that are actually bankrupt, and recall, which measures the proportion of actual bankrupt companies that are correctly identified, provide a more meaningful evaluation of a model's ability to detect the minority class. Precision-Recall analysis is therefore particularly well suited to this project, as it focuses directly on the trade-off between correctly identifying bankrupt companies and avoiding false bankruptcy predictions.

The main objective of this project is to design and evaluate a machine-learning-based system that predicts the bankruptcy status of a company using its financial indicators, with particular attention to handling class imbalance and evaluating performance using precision-oriented metrics. The general workflow of the proposed system involves collection of a financial dataset, data preprocessing, training of the Decision Tree and Random Forest models, and evaluation of their predictions on a held-out test set. Ultimately, this project compares the Decision Tree and Random Forest models to identify which model provides the best balance between detecting bankrupt companies and avoiding incorrect bankruptcy predictions.

3. SOFTWARE REQUIREMENTS:

Operating System:

Windows 10 or Windows 11

Programming Language:

Python 3.x

Development Environment:

Jupyter Notebook or Google Colab

Required Software and Python Libraries:

- Python
- Jupyter Notebook or Google Colab
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn
- Imbalanced-learn

- Joblib

Machine-Learning Algorithms:

- Decision Tree Classifier
- Random Forest Classifier

Evaluation Methods:

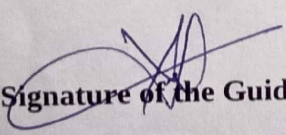
- Confusion Matrix
- Precision
- Recall
- F1-Score
- Precision-Recall Curve
- Average Precision Score
- Classification Report

Dataset Requirement:

A CSV dataset containing company financial indicators and a target column indicating whether a company is bankrupt or non-bankrupt.

Minimum Hardware Requirements:

- Intel Core i3 processor or equivalent
- Minimum 4 GB RAM
- 8 GB RAM recommended
- At least 1 GB of available storage
- Internet connection for Google Colab, downloading the dataset, and installing libraries


Signature of the Guide
Course Instructor